

3.2.3. Model Manipulations

We also explore a representation-based approach that intervenes directly on the model’s residual stream at generation time. The intervention operates in two stages: feature identification and steered generation.

Feature identification For each forbidden word w_i , we identify the SAE feature most strongly associated with it by running the model on the short natural prompt "*La parola vietata è w_i* " (translated: "*The forbidden word is w_i* ") and extracting the residual-stream activations at layer 29 of `google/gemma-3-4b-it`². These activations are passed through a pre-trained JumpReLU SAE from `google/gemma-scope-2-4b-it` [20], using a residual-post hook at width 65k. The feature with the highest activation across the token span corresponding to w_i is selected as its representative. This process is applied to all forbidden words in a card, yielding a set of feature indices $\{f_1, \dots, f_k\}$.

Steered generation At generation time, we register a forward hook on the same residual-stream layer. At each decoding step, we compute the mean decoder vector $d = \frac{1}{k} \sum_j \mathbf{w}_{f_i}^{\text{dec}}$ across the selected features and apply an activation-norm-scaled intervention:

$$\mathbf{h}' = \mathbf{h} + \alpha \cdot \|\mathbf{h}\|_2 \cdot \mathbf{d}$$

steering the residual stream *away* from the direction associated with the forbidden concepts. The model is otherwise prompted without any mention of forbidden words, receiving only the target word and the standard output instructions. The entire process operates at inference time without modifying model weights.

3.3. Evaluation

Exact amount of violations We evaluate constraint compliance in the description generation through a number of metrics. *Accuracy* measures the percentage of valid descriptions that fully comply with the rules of the game, i.e. outputs that contain neither verbatim forbidden words nor any morphologically related form. *Share of Exact Violations* reports the percentage of descriptions in which the model uses one of the forbidden words verbatim. *Share of Morphological Violations* reports the percentage of descriptions in which the model produces a word sharing the same stem or lemma as a forbidden word³. *Share of no <eot>* applies exclusively to `openai/gpt-oss-20b` in reasoning mode and it measures the proportion of cases in which the model exhausts its 2048-token reasoning budget without producing a final description. Finally, as anticipated in Section 3.2.1, the prompt does not explicitly instruct the model to avoid using the target word itself in the description. *Share of Target Leaked Words* reports the percentage of outputs in which the model nevertheless commits this error, inadvertently revealing the answer it was supposed to help guess.

²Layer 29 corresponds to $\sim 85\%$ network depth, where representations are richer in semantic content (see Appendix C).

³Detected using the `it_core_news_sm` spaCy model [22] and the `nltk` Italian Snowball stemmer [23].

LLM-as-a-judge To assess the communicative effectiveness of the generated descriptions, we employ an LLM-as-a-judge evaluation [24] as a complement to human evaluation. We use both `google/gemma-3-4b-it` and `openai/gpt-oss-20b` as automatic judges, evaluating descriptions generated by each model with both judges — that is, each model judges its own outputs as well as those of the other. Each judge is presented with the generated description alone, with no access to the forbidden words or the target word, and asked to guess the word being described. Instead of generating a free-form answer, we extract the judge’s next-token probability distribution immediately after reading the description and measure how much probability mass is assigned to the first token of the target word. This allows for a fine-grained, continuous assessment of how strongly the description evokes the target concept, rather than a binary correct/incorrect judgment.

We derive two metrics from this distribution: *Pass@k* measures whether the target word’s first token falls within the top- k most likely tokens, $k \in \{1, 2, 3, 5, 10\}$; the *Raw token likelihood* reports the actual probability assigned to the target.

Human Evaluation We conduct a human evaluation study involving 8 annotators, structured into two phases on disjoint sets of cards.

In the **guessing phase** (80 cards), annotators are presented with model-generated descriptions and asked to guess the target word as if playing an actual game of Taboo, albeit without the interactive component. Each annotator sees 20 cards, with system conditions distributed randomly across annotators to ensure balanced coverage of all evaluated systems, and each card evaluated under two different conditions by two different annotators.

In the **description phase** (100 cards, disjoint from the guessing phase), each annotator receives 20 cards. For 5 of these, they produce a spontaneous, oral-style description; for the other 5, they write a deliberate, carefully worded description. This design mirrors the *without thinking* and *with thinking* regimes observed in model experiments respectively. These 10 descriptions of unique items produced by each annotator are then presented to the other annotators for guessing, with the oral description setting being the closest to the actual Taboo game. The last 10 cards are held constant across all annotators: every participant describes the same items, enabling future analysis of inter-annotator variability in description strategies, which we leave for future work.

4. Results

4.1. Models are quite good at following given constraints

Figure 2 provides an overview of model performance across forbidding experiments, compared against a *baseline* in which models generate target word descriptions without any forbidden word constraint.⁴ Baseline accuracy varies considerably across models, ranging from 24.7% for `openai/gpt-oss-20b` without reasoning to 34.0% for `google/gemma-3-4b-it`, reflecting how readily each model would spontaneously use the forbidden words for descriptive purposes, even without knowing they are banned.

⁴Prompt in Appendix A; full results in Tables 4–6 in Appendix E.

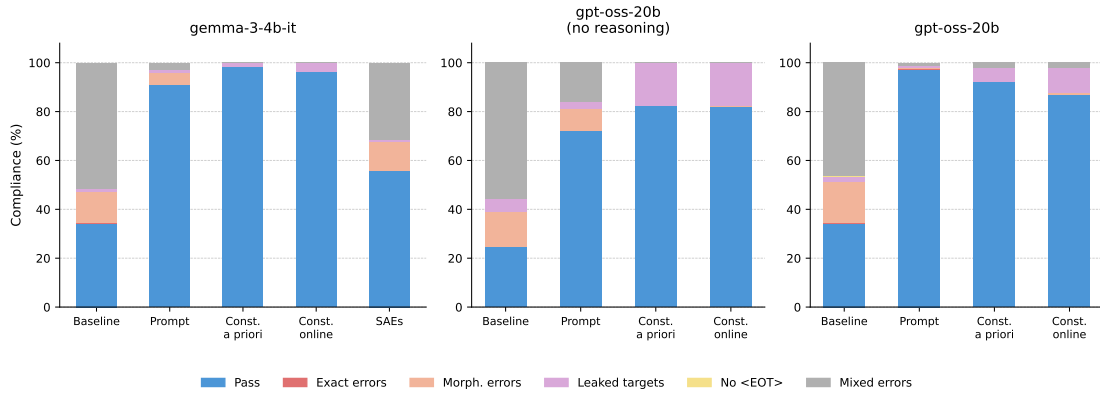


Figure 2: Compliance breakdown by method and model. Bars are partitioned into mutually exclusive categories summing to 100%; outputs with multiple error types are grouped into Mixed. The Baseline frequently violates taboo rules, while prompting substantially reduces errors. Constrained decoding achieves near-perfect compliance by construction, with residual failures almost exclusively due to target-word leaks. For gpt-oss-20b, enabling reasoning further improves compliance.

Under the prompting constraint, accuracy rises markedly for both google/gemma-3-4b-it (91.2%) and openai/gpt-oss-20b with reasoning (97.4%). For google/gemma-3-4b-it, a small proportion of exact and morphological violations persist; openai/gpt-oss-20b with reasoning exhibits only morphological violations, suggesting that chain-of-thought is effective at suppressing verbatim forbidden words but less reliable on morphologically related forms. When reasoning is disabled, accuracy drops to 72.2%, with a higher share of both violation types, confirming that the reasoning process plays an active role in enforcing lexical constraints.

In the constrained generation setting, all three models achieve high accuracy, with google/gemma-3-4b-it reaching near-perfect compliance under both a priori (98.5%) and online (96.4%) settings. openai/gpt-oss-20b with reasoning performs similarly under a priori constraints (92.3%), though accuracy drops in the online condition (87.1%), accompanied by a higher rate of target word leaks. Without reasoning, the model’s accuracy reaches 82.5% and 82.0% respectively, with a markedly higher proportion of leaked target words in both cases.

Finally, the SAE-based approach (Section 2), evaluated on google/gemma-3-4b-it only, reaches 55.7%, well below both prompting (91.2%) and constrained generation (98.5% and 96.4%), confirming that representation-based interventions alone cannot match explicit lexical guidance, though the result remains well above the unconstrained baseline (34.0%).

4.2. The type of constraining strategy shapes description informativeness

Having established that constraining strategy shapes compliance, we now ask whether it also affects how informative the resulting descriptions are as measured by the ability of a judge model to identify the target word.

The two judge models agree on the broad ordering of conditions but differ substantially in sensitivity: openai/gpt-oss-20b assigns near-zero probability to the target word across almost all conditions, with pass@10 never exceeding 10%; google/gemma-3-4b-it is consistently

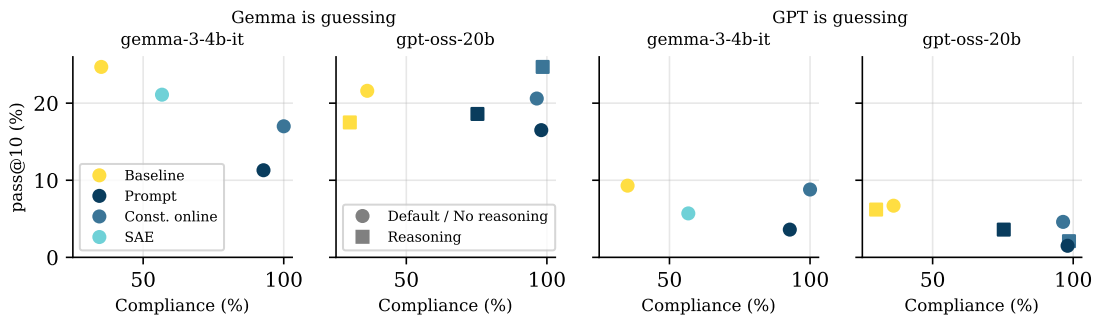


Figure 3: Compliance vs. pass@10 across models, methods, and evaluators. All three proposed methods (Prompt, Constrained, SAE) substantially increase compliance over the Baseline, confirming their effectiveness at enforcing the Taboo constraint. Pass@10 remains broadly stable across methods, suggesting compliance gains do not come at the cost of description quality. Gemma, in the guesser role, consistently achieves higher pass@10 than GPT both in- and out-of-distribution. Finally, enabling reasoning in gpt-oss-20b does not yield consistent improvements in its description effectiveness.

more generous, reaching up to 24.7% on the same descriptions (Figure 3).

Despite this gap, both judges agree that the constraining strategy matters. Descriptions produced under inference-time constrained generation are in most conditions easier to guess than prompt-based ones: for `google/gemma-3-4b-it` as clue-giver, constrained online generation yields pass@10 of 17.0% and 8.8% against 11.3% and 3.6% for prompting under the `google/gemma-3-4b-it` and `openai/gpt-oss-20b` judge respectively. A likely explanation is that forcing the model away from forbidden tokens at decoding time pushes it toward less obvious but more semantically precise paraphrases, whereas an instructed model tends toward overly cautious, informationally sparse descriptions.

SAE-based steering presents the most striking dissociation: compliance is low (56.7% strict accuracy) yet communicative effectiveness is competitive, with pass@10 reaching 21.1% under the `google/gemma-3-4b-it` judge, also exceeding constrained generation for the same clue-giver. This suggests that blocking surface tokens at decoding time imposes a descriptive cost that representation-level intervention avoids: the model loses access to forbidden words but retains the underlying conceptual associations needed to produce an informative description.

4.3. Models are weak guessers

We generated descriptions for 10 shared Taboo cards using `google/gemma-3-4b-it`, `openai/gpt-oss-20b` (with and without reasoning), Claude Sonnet-4.6 (max effort) [25], Gemini-3.1 Pro [26]⁵, along with human-generated ones. These are passed to `openai/gpt-oss-20b` and `google/gemma-3-4b-it` in order to be guessed and the results are reported in Figure 4. `openai/gpt-oss-20b` assigns near-zero probability to the correct target across virtually all clue givers, reaching at most 1.2% at pass@10 on human descriptions. `google/gemma-3-4b-it` is substantially more capable, yet still requires between 5 and 10 guesses to approach a decent accuracy: on human-written descriptions, its pass@1

⁵Proprietary models are used in their chat version.